The Impact of Tariffs on China Dairy Production System: The Case of Twin Tariffs

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Abstract

This paper develops the "Twin Tariffs" theory, where China imposed tariffs on feed inputs and dairy products. Using a time series growth response model with fixed effects and a large country twin tariffs equilibrium model, we quantify the tariff effects on input costs, import volumes, and farm-gate milk prices. Twin tariffs curtailed feed import quantities, which in turn put upward pressure on Chinese dairy input costs, and small farms face a higher risk of being squeezed out. In the long term, tariffs on alfalfa and dairy have opposite impacts on domestic milk products price, as dairy imports are more elastic than alfalfa in China. Chinese domestic milk products prices rose by 9% during the initial trade war period (2018-22) due to the higher cost of alfalfa. Following tariff adjustments and feed subsidy policies, China's farm-gate milk prices have begun to stabilize. Our findings carry important policy implications: tariff-induced cost shocks in a large importer country, suggesting that trade policy should carefully weigh protectionist measures while considering critical food supply chains and social welfare changes.

Keywords: Tariffs; Alfalfa; Dairy; Trade War; Vertical Supply Chain

JEL Classification: F13; Q17; Q18

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Introduction

Tariffs are tools used by nations to address domestic political and economic objectives. However, majority of the literature in agricultural economics are analyzing this issue through the lens of single-commodity impacts. The 2018 U.S. China trade war introduced a more complex phenomenon, in which we define as "twin tariffs"- the simultaneous imposition of duties on inputs and output products. China enacted retaliatory tariffs of more than 25% on agricultural goods from the U.S., which affected 86.3% HS 6-digit products in 2018 due trade war (Chor & Li, 2024; P. D. Fajgelbaum & Khandelwal, 2022). China lacks a comparative advantage in land-intensive feed grains and protein sources such as alfalfa and soybean. So alfalfa production in China cannot meet the demands of growing animal husbandry in both yield and quality (Wan et al., 2022). China's demand for alfalfa has grown rapidly, imports supplied about 49% of total demand in 2015 (Wan et al., 2022). By 2023, the U.S. accounted for about 89.9% of China's alfalfa hay import volume (Hou et al., 2024). In the end of 2023, China's State Council Tariff Commission announced an extension of the Section 301 retaliatory tariff exclusions on 12 agricultural products, including alfalfa hay, through July 31, 2024 (Qian, 2023). After several rounds of countermeasures, tariffs on more than 90% of the U.S. agricultural exports to China reaching 25% on the U.S. exports like alfalfa and soybeans, which has significantly affected the cost of dairy feed (P. Fajgelbaum et al., 2024; Xu et al., 2025). The heightened tariffs led to increased cost of imported alfalfa, which are vital inputs for dairy farming. Policymakers should navigate the trade-offs between protecting domestic industries and fostering an open, liberalized trade environment that enhances global food systems (Interesse, 2024).

The central question arising from this situation is how these "twin tariffs" [1]. Free trade policies are widely viewed as contributing to improved food security by enabling countries to exploit their comparative advantages, reducing food insecurity by increasing a country's gross domestic product, per capita incomes, and funding for social safety net programs (Adenäuer et al., 2025; Smith & Glauber, 2020). Richman et al. (2014) generally sees tariffs as a legitimate tool to address chronic

[1]Twin tariff: tariffs imposed on both inputs (alfalfa) and outputs (dairy), tariffs imposed on alfalfa increased the dairy inputs cost, then the dairy supply curve shifts to the left.

trade imbalances, as the Scaled Tariff (ST) brought trade toward balance, tariff revenue would be replaced by increased revenue from higher American incomes. Conceptually, long-run changes to trade barriers likely affect not only farmers and producers in a region but also with consumption and even food security. The theoretical foundation posits tariffs distort market efficiencies and allocation of resources and highlights changes in productivity, input prices, and consumption patterns (Fertő et al., 2021; Smith & Glauber, 2020). China's demand of imported feed stocks, including alfalfa and soybeans underscores the vulnerability of its dairy sector to external shocks. Bu and Liao (2022) noted that China's rural revitalization initiatives aim to reduce reliance on imports by incentivizing domestic forage production. However, structural challenges, such as aging populations and high production costs, continue to hinder progress. Tariffs generally insulate producers from competitors, but their costs to domestic consumers can be an incentive for unilateral liberalization and loss of social welfare (Farrokhi & Pellegrina, 2023; Gaigné & Gouel, 2022). Subsidize domestic agricultural production as a mitigate measure that expand domestic production and exports while reducing imports, is used in China during trade war (Farrokhi & Pellegrina, 2023).

This study provides a novel analysis of the trade war's impact by focusing on a previously unexplored aspect – the simultaneous tariff shock on both input and output market of dairy industry. Besides, this study estimates the welfare changes under different tariffs scenarios (to compare results of single tariff, twin tariff, free trade and prohibit trade). Using time series data of dairy and alfalfa trade between China and U.S., and employing fixed-effects modeling, we isolate the effects of the 2018–2020 tariff increases on dairy production, feed imports, and farm input costs. Our approach captures the cascading effects of an input cost shock coupled with an output demand shock. The results illustrate how the twin tariffs impact Chinese dairy production and altered import patterns, while also indirectly affecting the U.S. dairy markets through supply and price fluctuations (as evidenced by the decline in the U.S. milk prices during the dispute). By shedding light on these dynamics, our paper contributes to the literature on trade policy and food systems in several ways. First, it offers a case study of how trade barriers on a feed commodity (alfalfa) ripple through to influence livestock output (dairy), highlighting the vulnerability of interlinked markets (Fertő

et al., 2021; Koppenberg & Hirsch, 2022). Second, it provides empirical evidence from the Chinese context, enriching the understanding of how a major importer responds to sudden trade disruptions. Finally, the insights from this twin tariff case can inform global trade discussions, serving as a reference for other countries contemplating tariffs in sectors where imported inputs are crucial for domestic food production.

Background

China's rising demand for dairy products has made the tariff issue increasingly important, as shifting dietary patterns place greater emphasis on nutrient-rich foods, especially protein and calcium derived from milk (Gooch, 2017). Per capita dairy consumption in China has been on a steady upward trajectory, driven by urbanization, income growth, and government nutrition campaigns that highlight dairy's role in improving public health. However, domestic production has struggled to keep pace, particularly in the Golden Milk Source Belt, Inner Mongolia, where rising feed costs and land constraints hinder output growth.

Alfalfa is the 3rd. most valuable cash crop in the U.S. and more than 50% of export forge is alfalfa (see appendix figure A7, A3), with an estimated economic value of $8.7 billion in 2023 (Cantor et al., 2022; Felxi, 2024). Alfalfa is the third valuable field crop and planted about 23 million acres each year in the U.S. It is also considered one of the most important forage in the world because of its high nutritional value for livestock (Cantor et al., 2022). So alfalfa is a vital export cash crop in U.S. Consequently, tariffs imposed on both dairy and its critical inputs exacerbate cost pressures, undermine food security, and complicate China's efforts to meet its population's nutritional needs. The figure 1 demonstrated the dynamic impacts of the twin tariffs on the dairy supply chains in both countries. Dairy production declined from 2018 to 2020 due to increased input costs and tariffs restrictions in China. Following China's imposition of tariffs on primary feed ingredients and dairy products, the price of whole milk in the U.S. decreased, leading to varying degrees of economic losses across states because of retaliatory tariffs.

China introduced producer subsidies for local forage growers and expanded support for regional

dairy production to mitigate cost shocks and stabilize farm-gate milk prices. Besides, China made efforts in trade diversification away from the U.S. PIIE research shows that China has sought out alternative markets for its agriculture imports, even while the U.S. remains reliant on Chinese import demand (Qian, 2023; Rose, 2024). China has also ratified its participation in the Regional Comprehensive Economic Partnership (RCEP), and applied to join the Comprehensive and Progressive Trans-Pacific Partnership (CPTPP) (Li et al., 2020; Rose, 2024). China's diversification into alternative markets may push future tariffs impacts to be smaller bite and weaker leverage. China's imports of alfalfa meal/pellets totaled 15.12 million kg in 2023, primarily sourced from Italy and Spain, with only 24,745 kg imported from the U.S., representing a decline of about 90% compared to 2022. However, the U.S. remained China's principal supplier of imported alfalfa hay, providing 898,186 tons, accounting for 89.9% of the market (appendix figure A5). In 2024, China imported $1.42B of milk, becoming the 2nd largest importer of milk in the world. The U.S. is the third largest supplier of dairy products to China which include whey, milk powder, and lactose. These are all primary processed milk products. Therefore, examining the empirical correlation between twin tariffs and domestic milk prices is important to reveal how alternative tariff scenarios may shape market outcomes and to provid policy intuition.

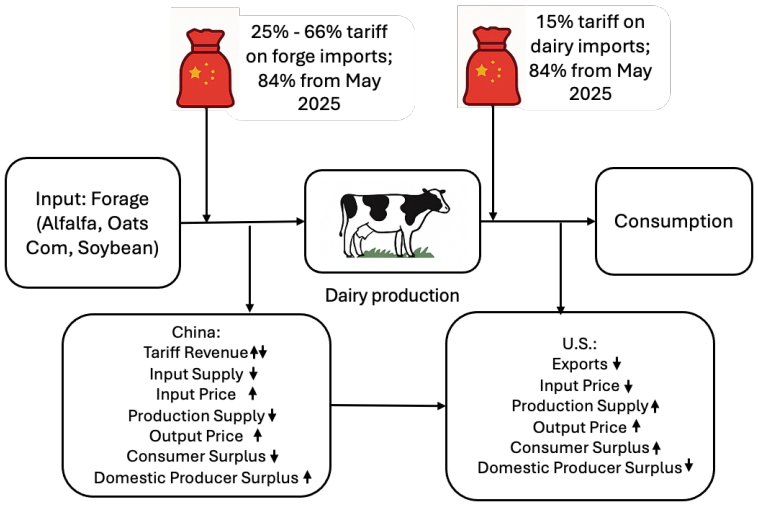

Figure 1: Framework of Chinese tariff on U.S. alfalfa and dairy impacts

Conceptual Framework

Large country single commodity trade

Trade barriers distort resource allocation, impacting not only producers but also consumers and food security. The U.S. remains a key supplier of alfalfa, which is a critical component of dairy feed. Tariff barriers have led to increased inputs costs for dairy production and thereby affecting productivity in China. Following China's imposition of tariffs on primary feed ingredients and dairy products, the price of whole milk in the U.S. decreased, leading to varying degrees of economic losses across states because of retaliatory tariffs. Chinese dairy price observed a significant increase, while production saw a slight decrease from 2018 to 2020 due to increased input costs and tariffs restrictions. Due to the rise in tariffs, the cost of forage increased, and domestic demand is partially reduced accordingly. In the short run, the supply of dairy cow feed decreased, and the average cost of dairy production rose in China. Chinese twin tariffs imposed on both alfalfa and dairy have complex impacts. This policy redistributes welfare across stakeholders: Chinese consumers lose surplus due to the higher price (a Harberger triangle representing deadweight loss), domestic producers gain surplus from the price increase, and the government collects tariff revenue (Schmitz et al., 2022). The imposition of tariffs disrupted dairy inputs supply chains, leading to increased costs for China's dairy production and reduced competitiveness for U.S. exporters (Smith & Glauber, 2020; Unveren & Luckstead, 2020).

As shown in figure 2, given the input supply S and demand D, the free trade price is P_f. For the tariff T_0 under the small country assumption, the restricted import volume is $Q' - D_s$. In the large country case, the tariff T results in a terms-of-trade effect where the world price is depressed to P_b. The resulting tariff revenue is $(ehgf)$ rather than $(cdef)$. (Schmitz et al., 2022). For example, The China import alfalfa CIF price from U.S. is $284.37 per ton in 2017, after the first round of tariff war, China imposed 20% tariff rate on alfalfa (284.37*1.2 = 355.46), but the real price in 2018 is $336.59 per ton. Dairy import price is $1491.44 per ton in 2017, theatrical price after tariff is $1789.72 per ton but real price is $1852.62 per ton in 2019, it indicated that twin tariff makes

worse than single tariff. Because China is a large importer, the reduction in its import demand also depresses the world price of the good, meaning foreign exporters absorb part of the tariff by receiving a lower export price. This terms-of-trade effect allows China to capture some surplus from foreign producers, partially offsetting the domestic welfare loss relative to a small-country case. By contrast, in a small-country scenario the world price remains unchanged (no foreign price concession) and the entire tariff burden falls on importers, so any import tariff yields a net welfare loss equal to the classic dead-weight loss areas. We extend this framework to China's "twin tariff" policy of concurrently taxing a crucial input (alfalfa feed) and the output (dairy products). A tariff on imported alfalfa raises dairy farmers' feed costs and changes the China's dairy supply curve (reflecting higher marginal costs), pushing up the farm-gate milk price.

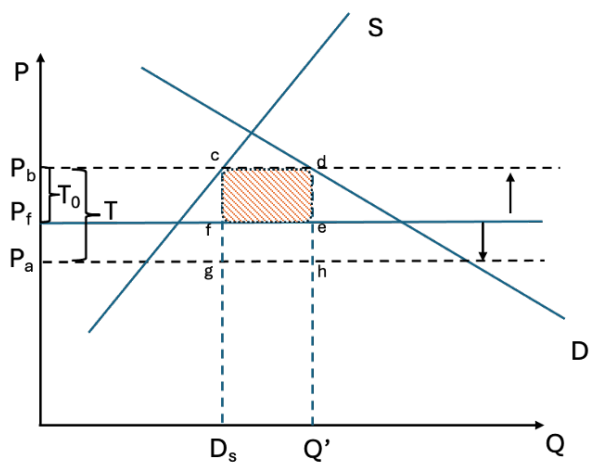

Figure 2: Effects of tariffs on terms of trade effects

Twin tariff model

The twin tariffs imposed on forage and dairy increased the dairy inputs cost, and supply prices due to trade war in China. The figure 3 use a simple model illustrated that input and output changes with twin tariffs on both of inputs and outputs by China. P1 is free trade price of alfalfa, Qd is domestic supply, (Fs – Qd) is import amount of alfalfa. After tariffs imposed on it, the price of alfalfa increased to P2 and import amount reduced to (Q2 – Q1), supply of alfalfa decreased to Q2.

The dairy import amount decreased from (Fs - Qd) to (Q4 – Q3) due to tariffs and price increased to P4. The tariffs collected by China government on alfalfa is (abcd) green rectangle and (efgh) blue rectangle on dairy production. Besides, the dairy supply curve shift to the left due to supply of alfalfa decreasing and marginal input cost increased, so the price of dairy increased significant. Higher tariffs on alfalfa and oat hay resulted in a price surge of 20–25% for these inputs (Rose, 2024).

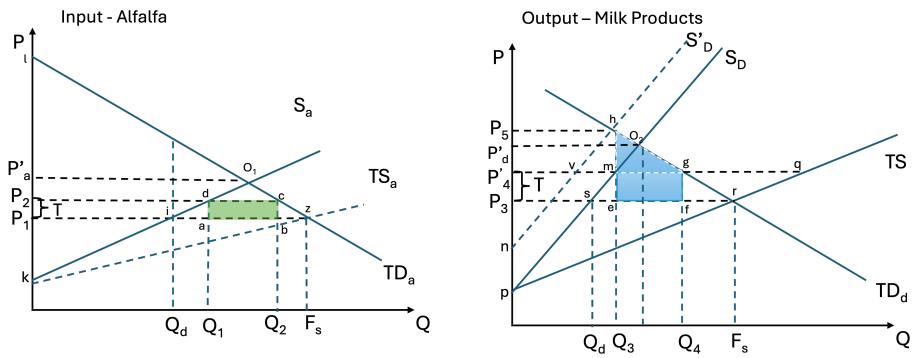

Figure 3: Twin tariffs impact on Chinese dairy industry

As a result, dairy farmers in Inner Mongolia (see appendix A1)faced increasing production costs, leading to a 10% rise in retail milk prices by 2021. But higher price of alfalfa hay simulates the productivity of Chinese domestic farmer. Before 2022, Chinese importers worried that potential future lockdowns would increase freight prices further and thus decided to expand and replenish their inventories. China imported 999,518 tons of alfalfa hay in 2023, a decrease of about 40% compared to 2022 (see appendix figure A3). According to the statistics of China Customs, alfalfa imports decreased 0.7% in the first quarter of 2024, but import value was US $164 million and declined 35.0%. The average CIF price of alfalfa peaked at $596.1/T, then declined to $383.60/T in 2024, driven by inventories and weakened demand in China, and preliminary data indicate an upward trend in 2025. The core theoretical argument is that twin tariffs are more distortional than a single tariff because they jointly affect production of inputs and outputs, with the magnitude of the effect shaped by alternative patterns and market elasticities.

Follow-up policy: producer subsidy effects

Chinese domestic milk products price shift higher due to twin tariffs. The Chinese want to stable both alfalfa and milk products price (P_1 and P_a in figure 4), so subsidies policy for local alfalfa forage growers implemented in 2020 to reduce input cost. As shown in figure 4, P_2 is equilibrium price after tariff imposed on alfalfa, but when subsidy imposed on alfalfa the price will shift to P_1. The milk products price will shift back due input price decrease, even there are twin tariffs.

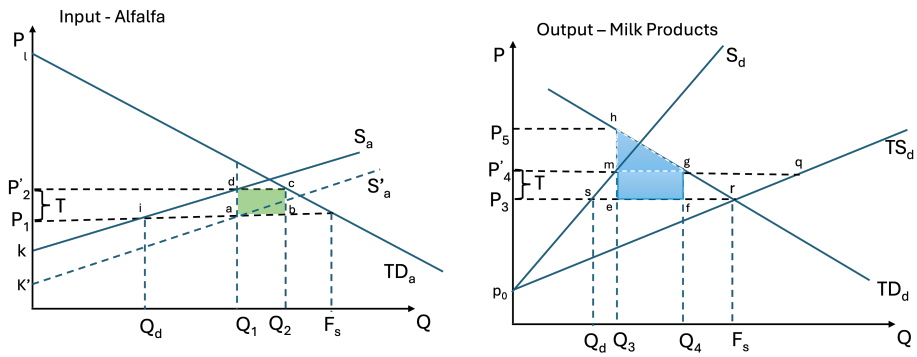

Figure 4: Implemented subsidies on alfalfa after tariff imposed

Following the Chinese government's implementation of a subsidy to alfalfa farmers as a countermeasure. The optimal subsidy provided to domestic alfalfa suppliers shifts the alfalfa supply curve from Sa to S'a, there will be no trade on alfalfa in extreme case period (other configuration would allow for trade alfalfa) (Kondaridze & Luckstead, 2023). Domestic alfalfa producer surplus will increase (k'kcd) and dairy producer surplus will increase (+P_4P_3em) relative to the unsubsidized scenario. Thus, alfalfa farmers are effectively protected without negatively affecting dairy producers, and consumer losses are mitigated compared to the twin tariffs scenario. In 2023, China produced 41.97 million tons of milk, an increase of 6.7% compared to 2022, marking the fourth consecutive year of a production increase of over 6% (Steinberg & Tan, 2024; Victoria, 2024). Additionally, the alfalfa hay import price decreased from $596/tonto400$/ton from 2022 to 2023 due to less demand of alfalfa import in China ("USDA ERS - Agricultural Trade," 2024).

Empirical Framework

Data

This study combines multiple data sources to analyze the impacts of twin tariffs on China's dairy production system. The dataset is a panel constructed from USDA FAS – Global Agricultural Trade System (GATS), China Customs Statistics (GACC), USDA ERS, and the U.S. Census Bureau International Trade Data, covering the monthly data from 2005 to 2025. These data provide monthly import values, quantities for alfalfa hay (HS 1214) and major dairy products (HS 04), which represent the primary traded inputs and outputs in the dairy supply chain. Data from USDA including China imports of alfalfa and dairy quantity and value from U.S., which used to check changes of China imports. Data from GACC including Chinese domestic farm-gate milk price, which used for estimate tariff rates impacts on domestic dairy production cost. From these, unit import values (USD per ton) were calculated as value-to-quantity ratios to proxy for trade prices. Tariff rates are collected according to the statistics of China Customs, figure 5 shows the Chinese twin tariffs fluctuation from 2010 to 2025. The appendix B table C1 shows overview of the data set. The China farm-gate milk price and the world dairy price index series were collected from Food and Agriculture Organization (FAO), which composite cost indices for the milk and dairy basket were used to track input–output price transmission.

Forage tariffs are uniform across the relevant HS6 products, while dairy tariffs vary across HS6 lines. However, most high-variation products, such as butter, lactose, and cheese, represent only a small share of Chinese consumption; the main imported dairy products (milk powder, liquid milk, and whey) face the same tariff rate. Together, these institutional features support the use of the twin-tariff framework as a reasonable approximation. This approach provides a consistent empirical foundation to examine how tariffs on both inputs and outputs—the twin tariff scenario—alter production costs, trade flows, and welfare outcomes. The empirical strategy integrates both descriptive trend analysis and econometric modeling. China imports of alfalfa and dairy price and quantity changes are shown in appendix figure A4, import dairy price increased and

quantity decreased significantly during the first round of trade war (2019-2020), so this study will estimate interrupted time series tariff shock effects of interested periods, pre-trade war (2017-2018) as a baseline period. Percent changes relative to pre-tariff baselines quantify the magnitude of trade disruptions.

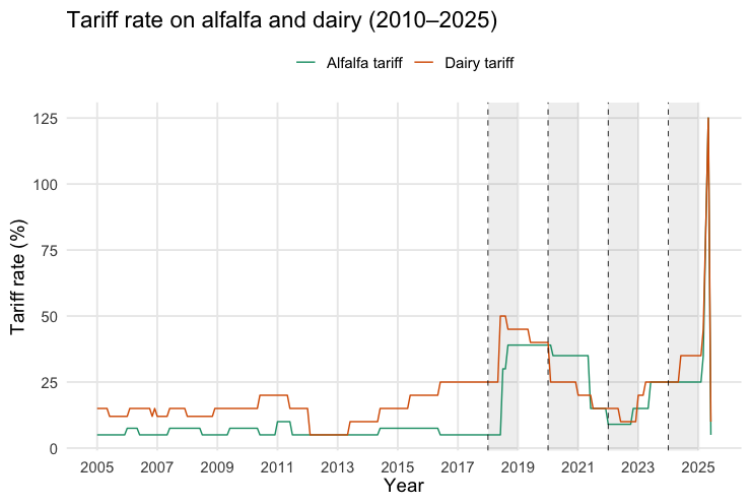

Figure 5: Chinese tariffs rate on U.S. dairy and alfalfa

Assumptions

In the large-country trade framework, China is modeled as a large importer of alfalfa and dairy products, while the U.S. is treated as the primary alfalfa and dairy exporter. It is implying its tariff policies influence not only domestic prices but also global prices. Thus, when China imposes tariffs, part of the burden is shifted back onto foreign exporters through price concessions, consistent with the terms-of-trade effect (Grant et al., 2021; Smith & Glauber, 2020). The twin tariff case was simulated using partial equilibrium diagrams to illustrate the compounding effects on dairy supply chains (Schmitz et al., 2022). By combining empirical evidence with theoretical modeling, this approach captures both the immediate quantity-price adjustments and welfare implications of China's tariff policies in the dairy sector (P. Fajgelbaum et al., 2024; Villoria, 2026).

Key assumptions in this application: 1. China as a large-country importer: China's import demand for alfalfa and dairy is dominant in global trade, accounting for the largest share of alfalfa

imports and 3rd largest share of dairy from United States. Consequently, tariff impositions in China are assumed to lower the world price. 2. Other exporters are small: Competing input suppliers (e.g., Spain, Italy, or Australia) are treated as small countries and price taker. Their market shares are marginal relative to U.S. exports, so the third-country effects are ignored. 3. Imperfect substitutes: Alfalfa is modeled as a critical and non-substitutable forage input in China's dairy production. While some substitution across forage crops (e.g., corn silage, oats) is possible in practice, this framework assumes inelastic substitution to isolate the impact of alfalfa tariffs.The United States is a major supplier of whey and milk powder to China, both of which are important processed products linked to farm-gate liquid milk, whereas China's liquid milk imports originate mainly from New Zealand. Therefore, when China imports less dairy from the United States, it may increase imports of liquid milk from other countries, which can subsequently influence the domestic farm-gate milk price. 4. Stable trend: Alfalfa is seasonal agricultural product, this study assumes import trends are similar across different years, so all impacts come from exogenous variables. We fit the time series data in auto-regressive integrated moving average (ARIMA) checked the assumption. As shown in appendix A13, the forecast trend accuracy reached 99%, which aligned the test data well. Use Δlog formation can remove undeserved trends (e.g. inflation and CPI), because it expresses percentage change responses to tariff shocks in a way that is statistically robust (stationary) and economically interpretable (elasticities).

Regression model

The twin tariffs' on dairy (output) and on alfalfa (input) create simultaneous supply and demand side shocks to the Chinese dairy production system. Import quantities respond strongly and immediately to tariffs, while domestic milk prices adjust with a delay as feed costs and dairy import availability change. The retaliatory tariffs were imposed as part of geopolitical negotiations rather than in response to dairy market fundamentals. Following Amiti et al. (2019), Chor and Li (2024), and Koppenberg and Hirsch (2022), we treat tariff variation as exogenous variables with respect to shocks on import quantities and prices. Import quantity is an endogenous mediator between tariffs

and domestic milk price. Tariffs directly affect import quantities, and import quantities in turn affect domestic milk supply conditions and prices. Because dairy imports may endogenously respond to domestic prices, we instrument $\Delta \ln Q_{D,t}$ with lagged tariff changes, which isolated the impacts pathway from tariff to domestic milk production and price. Lag structures and fixed effects account for dynamic adjustment and confounding shocks. The empirical framework focuses on the time-series adjustment of aggregate import quantities and domestic prices to tariff shocks, which is more appropriate for identifying input-cost transmission and farm-gate price effects (Jeff Luckstead & Stephen Devadoss, 2025).

With above assumptions, this study used time series growth response model with fixed effects to identify elasticities to tariff shocks and lag periods effects of tariff rate changes on import and domestic milk price. We treat the tariff changes as exogenous and assume that are uncorrelated with unobserved shocks to import value (Amiti et al., 2019; Nunn & Trefler, 2010). In this way, we can capture month-to-month variations caused by tariff shocks. Appendix table B1 reports the reduced-form results in the baseline model and augments the specification with a twin-tariff interaction term in the interaction model to test for compounding effects. While the individual tariff rates are insignificant in the baseline model, the interaction term is positive and statistically significant, indicating that the two tariffs compound effects. Moreover, the tariffs shock appear to be largely excluded from any direct effect on domestic milk prices, which supports its role as valid external instruments for identifying the twin-tariff transmission mechanism (Bohara & Kaempfer, 1991). This evidence suggests that the tariff shock operates mainly through the import supply channel, ultimately affecting domestic producer prices through shifts in imported quantities. Hence, the tariff variables satisfy the core instrument variables (IV) requirements of relevance and exclusion.

Log-linear regressions were estimated to test the sensitivity of import quantities and domestic milk price to tariff rates, with fixed effects capturing the tariff shock. The model included lag of tariff shocks on the right-hand side by 2 periods to accommodate lagged effects (Chor & Li, 2024). $\Delta \ln Q_{d,it}$ = log-difference of dairy import quantity (tons); $\Delta \ln Q_{a,it}$ = log-difference of alfalfa

import quantity (tons); $\Delta \ln P_{it}^{milk}$ = log-difference of farm-gate milk price; $\Delta \ln \mathrm{tr}_{i,t-k}^{A}$ and $\Delta \ln \mathrm{tr}_{i,t-k}^{D}$ = k-lagged log-differences of alfalfa and dairy tariff rates, respectively. μ_m are monthly fixed effects that absorb seasonal patterns (e.g. Chinese New Year, production cycles), while year fixed effects (λ_y) absorb macroeconomic shocks (e.g. inflation, recovery from melamine crisis). $u_t, \varepsilon_t, \nu_t$ = idiosyncratic error terms. The parameters γ in function (1) represent the import dairy production elasticities to tariff changes. The parameter β in (2) identify the impacts of the log-difference of the farm-gate milk price ($\Delta \ln P_{it}^{milk}$) with respect to the instrumented dairy import quantity and other price variables. The θ_k in (3) are the effects of tariff rate changes on farm-gate milk price. $\mathbf{X}_t$ is a vector of control variables includes FAO dairy index, fuel price (transport cost), COVID (dummy), import price of alfalfa and dairy.

$$\text{First stage:} \quad \Delta \ln Q_{D,t} = \alpha_0 + \sum_{k=0}^{2} \left(\gamma_k^D \, \Delta \ln(\mathrm{tr}_{t-k}^{D} + 1) + \gamma_k^A \, \Delta \ln(\mathrm{tr}_{t-k}^{A} + 1) \right) + \gamma' \mathbf{X}_t + \mu_m + \lambda_y + u_t, \tag{1}$$

$$\text{2SLS:} \quad \Delta \ln P_t^{milk} = \beta_0 + \sum_{k=0}^{2} \beta_{1,k} \, \Delta \widehat{\ln Q_{D,t-k}} + \beta' \mathbf{X}_t + \mu_m + \lambda_y + \varepsilon_t, \tag{2}$$

$$\text{Reduced form:} \quad \Delta P_t^{milk} = \theta_0 + \sum_{k=0}^{2} \left(\theta_k^A \, \Delta ln(\mathrm{tr}_{t-k}^{A} + 1) + \theta_k^D \, \Delta ln(\mathrm{tr}_{t-k}^{D} + 1) \right) + \theta' \mathbf{X}_t + \mu_m + \lambda_y + \nu_t. \tag{3}$$

Tariff revenue and social welfare

We perform a counterfactual analysis to calculate social welfare and tariff revenue under different tariff scenarios as following. P_i^{*0} is the pre-tariff benchmark CIF price and P_i^{*1} the post-tariff CIF price. $P_i = (1+\tau_i)P_i^*$ is import price after tariff., V_i is the trade value, and Q_i is the import volume. TR denotes the tariff revenue, and the deadweight loss (DWL) is computed from the domestic price change (areas depending on demand and supply slopes or elasticities). TOT gain is terms-of-trade gain; domestic transfer is tariff-induced transfer from consumers to government/producers. M_i^1 is

post-tariff imports.

$$V_i = Q_i P_i^* \tag{4}$$

$$TR_i = \tau_i Q_i P_i^* \tag{5}$$

$$TR = TR_a + TR_d \;=\; \text{TOT gain} \;+\; \text{domestic transfer} \tag{6}$$

Welfare change:

$$\Delta W = \left(P_a^{*0} - P_a^{*1}\right) M_a^1 \;+\; \left(P_d^{*0} - P_d^{*1}\right) M_d^1 \;-\; \left(DWL_a \cdot P_a \;+\; DWL_d \cdot P_d\right) \tag{7}$$

The tariff revenue boost temporary and following decrease after retaliatory tariffs imposed on alfalfa and dairy (see appendix figure A2). The China import alfalfa from U.S. totaled 2.8 million tons in 2022 and decreased to 0.89 million tons in 2023 due tariff barriers (USDA-FAS, 2023; Wang & Zou, 2020). The new tariffs are measured affects more than $18 billion in trade coverage, as compared with total U.S. merchandise imports of $3.826 trillion in 2023 (Steinberg & Tan, 2024; Wolff, 2024). U.S. export to China is forecaster to be down by over 1 billion dollars from 2023 ("China," 2023; Hanrahan, 2024; Tucker et al., 2024). Tariffs can temporarily boost tariff revenue when the targeted good is inelastic, since import volumes do not immediately contract in proportion to the tariff increase. In the short run, this inelastic response allows tariff revenue to rise despite higher duties (Hanrahan, 2024). However, the increase is not sustained: over time, higher costs reduce import demand, especially for feed inputs such as alfalfa, where substitution and domestic policy adjustments occur. The result is a sharp decline in both import volumes and tariff revenue after the initial spike, consistent with the theory that tariff incidence on inelastic goods is temporary and erodes once importers adjust (Amiti et al., 2019).

Results

Twin tariff impacts

From 2018 to 2022, the average prices for alfalfa increased by 15%, per capita milk output in Inner Mongolia declined by 9.2%. The 2022 alfalfa hay price represented a 35.6% increase from 2021. Demand for dairy products has been increasing in China over the past few decades, but imports of dairy products from U.S. have declined since 2018 and continue to show a downward trend (see appendix figure A9-A12). On the other hand, the rise in input costs pushed the Chinese dairy farmlands facing higher pressure. Though there is a reduction in China import dairy from U.S., dairy imports demand gap can be alternative by New Zealand and EU. However, the Chinese alfalfa imports demand are mainly from U.S. Besides, China's retaliatory tariffs on alfalfa and soybeans led to an oversupply in the U.S. market leading to a decline in feed costs (Unveren & Luckstead, 2020). Meanwhile, the average price of milk decreased 8.5% from 2022 to 2024 in U.S. due to export decline (see appendix figure A12). The Chinese 2023 Dairy Cattle Industry and Technology Development Report predicts that raw milk production is expected to continue to increase while production costs remained high, but average raw milk prices declined, leading to reduced profitability for many dairy farms (Zhang & Abler, 2023).

Table 1 reports the three main specifications. Column (1) presents the import-quantity equation (Equation (1)), documenting how each tariff is associated with changes in China's dairy import volumes. Column (2) reports the reduced-form price equation (Equation (3)), estimating the total association between tariff changes and the farm-gate milk price across all transmission channels. Column (3) adds the contemporaneous interaction term (Equation (**??**)) to test whether the twin-tariff configuration generates a compounding price effect beyond the sum of the individual tariff elasticities. Column (4) shows twin tariff impacts on domastic farm-gate milk price through import dairy quantity.

Table 1: Twin Tariff Impacts on China's Dairy Production System (One-way Fixed Effects)

		(1)	(2)	(3)	(4)
		$\Delta\log(\textit{dairy import})$	$\Delta\log(\textit{milk price})$	$\Delta\log(\textit{milk price})$	$\Delta\log(\textit{milk price})$
		(First stage)	*(Reduced form)*	*(Interact term)*	*(2SLS)*
Tariff on dairy $\Delta\log(\cdot)$	$t=0$	-0.583	0.058	0.008	0.011
	$t-1$	0.853^{*}	-0.014	-0.007	-0.015^{*}
	$t-2$	1.186^{**}	-0.035	-0.004	-0.009^{**}
Tariff on alfalfa $\Delta\log(\cdot)$	$t=0$	-1.708^{***}	0.052	0.011^{**}	0.022^{***}
	$t-1$	-0.145	0.004	0.003	0.005
	$t-2$	-1.131^{***}	0.030	0.011	0.009^{***}
Interact tariff				0.071^{*}	
Observations		248	248	248	248
Adj. R^2		0.413	0.971	0.982	0.683

Notes: One-way fixed effects (month and year) included. Heteroskedasticity-robust standard errors. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Column 1 indicated that both tariffs are associated with reductions in dairy import quantities, with the alfalfa tariff ($\gamma_0^A = -1.708$, $p < 0.01$) showing a larger contemporaneous effect than the dairy tariff ($\gamma_0^D = -0.466$). The cumulative alfalfa coefficient ($\sum_k |\hat{\gamma}_k^A| = 2.984$) substantially exceeds the cumulative dairy coefficient ($|\sum_k \hat{\gamma}_k^D| = 0.573$), confirming the asymmetric substitutability structure. This pattern is consistent with the input-cost channel emphasized in this study, whereby tariffs on alfalfa raise production costs throughout the dairy supply chain. Column (2) reports the total co-movement between tariff changes and the farm-gate milk price across all transmission channels. The contemporaneous alfalfa coefficient ($\hat{\theta}_0^A = 0.052$) is positive but statistically insignificant,

and the lag coefficients ($\hat{\theta}_1^A = 0.004$, $\hat{\theta}_2^A = 0.03$) are positive, confirming the upward price pressure once feed-cost increases work through herd-management contracts. Dairy tariff coefficients are positive ($\hat{\theta}_0^D = 0.058$) on contemporaneous period and negative across lags (cumulative ≈ -0.049), reflecting trade diversion: the dairy tariff primarily redirects sourcing toward New Zealand and EU suppliers rather than raising domestic prices, which consistent with the high dairy import demand elasticity and with the precautionary-stockpiling mechanism described in the Empirical Model section: importers front-run the tariff, temporarily shift the price signal. Column (3) adds the contemporaneous interaction term (alfalfa tariff $*$ dairy tariff) to the reduced form. The estimated $\hat{\theta}_{AD} = 0.071$ is positive and significant at the 10% level, confirming the twin-tariff configuration is associated with a price movement approximately 7.1% larger than the sum of the individual tariff associations. The alfalfa coefficient at $t = 0$ in column (3) turns negative (-0.011^{**}) while the interaction absorbs the compounding, reflecting the non-linear joint effect. The very high adjusted R^2 (0.982) is driven by the month and year two-way fixed effects; tariff coefficients are estimated from residual within-period variation. In columns (4), the contemporaneous effect ($t = 0$) of the alfalfa tariff is 0.022% more pronounced than subsequent delayed effects. However, these effects turn negligible in longer periods, because dairy imports shift toward countries with lower tariffs and more competitive pricing. This trend creates a difficult environment for domestic dairy farms, as they must navigate rising input costs alongside stagnating milk prices. The negative association between the alfalfa tariff and dairy imports implies that higher alfalfa costs may raise farm-gate milk production costs and reshape import sourcing patterns. China reduced imports of U.S. dairy products (whey and milk powder), while relying more on alternative suppliers such as New Zealand, which has a strong cost advantage in liquid milk production. The import substitution effect for dairy imports is stronger than for alfalfa. This structural difference arises because liquid milk imports are primarily sourced from New Zealand under a zero-tariff policy, whereas around 90% of China's alfalfa imports from the U.S. Sum up twin tariffs effects of 2 periods, it indicating that 1% increase in the gross twin-tariff factor is associated with an approximately 0.023% ($\sum \hat{\beta}_1 * (\hat{\gamma}^A + \hat{\gamma}^D)$) increase in the farm-gate milk price. In appendix table B1, column (4) shows that the interaction between

the alfalfa tariff and the FAO dairy price index is positive, indicating that alfalfa cost shocks pass through more strongly to domestic milk prices when global dairy prices are high. This suggests that twin tariffs effects become more intensive when the alternative of low-cost imports weakens. Overall, domestic farm-gate milk prices experience an immediate spike following the tariff shock; however, this effect diminishes after one period and approaches zero by the second period. The tariffs does not have significant direct effects on farm-gate milk price in the long run (in column 2). So the consumer and producer swallowed most tariff cost. Impacts of tariffs on domestic milk price are decreasing as periods extending, so we analyzed the trade war period effects following to check long run effects.

Interrupted time series effects on domestic farm-gate milk price

We set pre-tariff period (2017-2018) observations as model the baseline , and calculated different periods' relative price changes miners contrast period changes as average interrupted time series effects. Appendix A15 shows that tariff shocks raise alfalfa CIF prices and farm-gate milk price after tariff shock. With limited short-run substitution and incomplete foreign price offsets, feed costs rise and push up the dairy sector's marginal cost, so the tariff shock effects on input (alfalfa) cost and domestic farm-gate milk (output) price are uniformly positive (see appendix figure A16 and A14). Domestic milk prices rose by 9% during the initial trade war period (2018-22) (figure 6), and logistic was tighten in 2020 due to COVID-19. Following tariff adjustments period and Chinese subsidy policies, China's relative milk prices have begun to stabilize. The adjust (2022–23) exhibits the less increased effects because China adjusted temporal tariff rate temporal, alfalfa subside and import tariff tax refund programs for agricultural business. By contrast, the second-round trade war (2023–25) yield much smaller effects than the first round (2018–22), due to higher domestic production and a series counter measures (subsidy and import alternative).

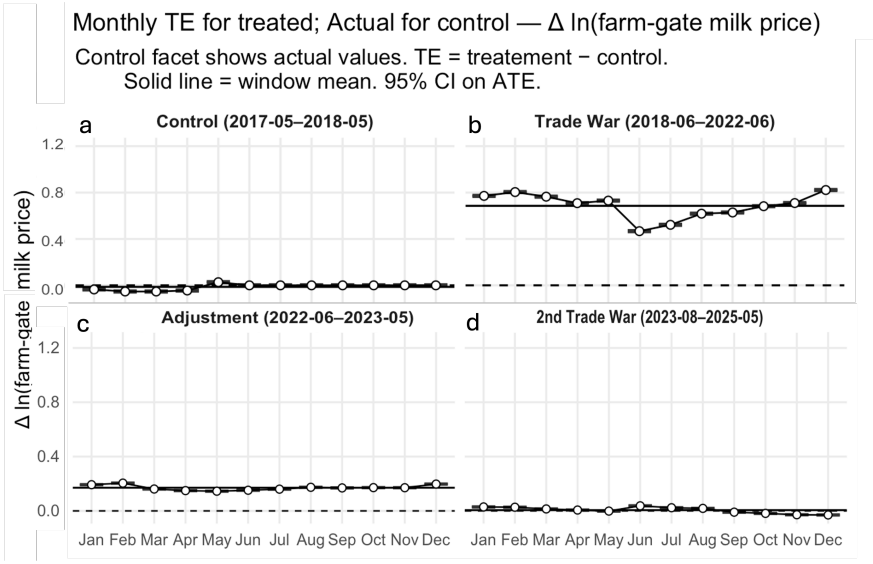

Figure 6: Interrupted time series tariff effects on China domestic farm-gate milk price

Importers and processors had time to adapt diversifying origins, expanding domestic forage, renegotiating contracts, and making selective use of tariff-exclusion channels—so the effective price impact per tariff point fell. This pattern suggests domestic producer subsidies and market segmentation, which prevent global trade policy shocks from destabilizing Chinese dairy farmers' returns. By the second trade war (2023-25) window, price effects attenuate, and import dairy and alfalfa quantity from U.S. decreasing, reflecting adaptation—tariff exclusions, renegotiated contracts, inventory smoothing, and diversification to non-U.S. origins and domestic forage. Both effects moderate further, consistent with a large-country terms-of-trade channel in which foreign exporters concede part of the tariff via lower prices while Chinese buyers improve elasticity over time. Retaliatory tariffs can boost tariff revenue in short term, but result in extraordinary welfare losses in long term. So, the cost of dairy production in China may continue increase in the future. Overall, the "twin-tariff" regime transferred the pressure to dairy farmer and increased the marginal cost of milk, with the strongest effects when global shipping and energy markets were tight.

In the short term, the tariff war has led to a higher increase in Chinese farm milk prices. However, milk producers (especially dairy farmers) face reduced profit margins because the proportionate rise in alfalfa feed prices exceeds the increase in dairy product prices from 2018 to 2022. Over the

long term, China's dairy import sources have diversified toward New Zealand, and the European Union, whose competitive pricing and product quality have intensified price pressure on domestic producers. As shown in appendix figure A6, the import alfalfa price increased 38%in 2022, but the farm-gate milk price increased less than 10% in this period (see appendix figure A11 and A14). Consequently, the twin tariffs impose short-run burdens on both dairy producers and consumers, but in the long run the pressure falls disproportionately on consumer.

Subsidy effects and Tariff Revenue

Different tariff scenario impacts on consumer, producer and tariff revenue are illustrated in table 2. We assume the 2017 price as the global price without tariffs, since it was one year before the tariff war and thus closer to the true world market level. According to the China Grass Industry Statistics (CGIS), China produced about 39.3 million tons of alfalfa hay in 2017 and 47.4 million tons in 2019, with only 0.18% exported. Because exports represent a very small share of total production, we do not consider them in the analysis. Government subsidies were implemented at the end of 2019, so we use 2020 prices to calculate subsidy effects. China's alfalfa imports account for about 50% of total demand, and in 2018, 96% of those imports came from the United States (Wang & Zou, 2020). If tariffs were imposed only on alfalfa, tariff revenue would decline, dairy producers would earn less, and consumers would lose more compared with the twin-tariff case. By contrast, if tariffs were imposed only on dairy products, alfalfa farmers would not incur losses, while dairy producers would experience relatively smaller gains, but overall social welfare losses would be greater. According to the USDA–FAS report, China produced 35.45 and 33 million tons of dairy in 2017 and 2019, respectively, consistent with our theoretical framework that higher tariffs decrease domestic dairy production by increasing input costs. The CIF price of U.S. alfalfa imports was $284.37 per ton in 2017. After the first round of tariffs, China imposed a 20% tariff on alfalfa (284.37×1.2 =355.46), but the observed price in 2018 was $336.59 per ton. For dairy, the import price was $1,491.44 per ton in 2017, with a theoretical post-tariff price of $1,789.72 per ton, while the actual observed price in 2019 was $1,852.62 per ton. This indicates that the combined (twin)

tariffs had a more severe impact than a single tariff alone.

Table 2: Impacts estimation of different scenarios of tariff imposed on alfalfa and dairy

Case scenario	Production surplus changes		Consumer surplus changes	Tariff revenue	Net loss
	Alfalfa	Dairy			
Twin tariff	+P1P2di (+$2263.7 M)	P5hn – P3sp (+$4878.2 M)	–P5P3rh (–$23035.9 M)	abcd + efgh ($185.9 M)	–$15708.1 M
Tariff on alfalfa	+P1P2di (+$2263.7 M)	P4vn – P3sp (+$1268.2 M)	–P'dP3rg (–$4011.2 M)	abcd ($64.6 M)	–$414.7 M
Tariff on dairy	0	+P3P4ms (+$10208.6 M)	–P4P3rg (–$19003.5 M)	efgm ($68 M)	–$8726.9 M
Prohibit tariff	+$P_1 P_a' O_1 i$ (+$2021.5 M)	P5hn – P3sp (+$4878.2 M)	–P5P3rh (–$23035.9 M)	No trade	–$15894 M
Subsidy on alfalfa	+k'kcd (+$3520.1 M)	+P'4P3sm (+$7487.8 M)	–P'4P3rg (–$13953.9 M)	efgm ($167.8 M)	–$2797.8 M

Tariff revenue is an endogenous fiscal outcome because both the tariff wedge and the import quantity adjust in equilibrium. Consistent with our comparative statics, retaliatory tariffs can generate a short-term increase in revenue when import demand is initially inelastic, but revenue declines as quantities contract and substitution/trade diversion intensifies (see figure A2). The subsidy on alfalfa counterfactual highlights this linkage: by shifting domestic alfalfa supply outward (figure 4), the policy compresses alfalfa import demand and therefore reduces the scope for collecting alfalfa-tariff revenue, effectively replacing fiscal receipts with higher domestic surplus in the upstream feed market. At the same time, tariff revenue from the remaining taxed segment (dairy imports: $efgm$) can persist (table 2 reports $efgm = \$167.8$M), while total welfare losses fall sharply relative to the twin-tariff benchmark (net loss $-\$2{,}797.8$M vs. $-\$15{,}708.1$M). This highlights an important policy implication: maximizing tariff revenue is not equivalent to maximizing welfare, and a subsidy that relaxes the binding feed-cost channel can improve aggregate outcomes even if it reduces the government's short-run tariff revenue.

Importing companies pay tariffs and decide whether to pass on any portion of the cost to consumers through higher prices in China (Schmitz et al., 2022). The tariffs that China imposed on U.S. exports were calculated to broadly match the value of U.S. tariffs, avoiding a spiral of escalation, but they were economically costly for the U.S. agricultural exports to China suffered, compelling U.S. to channel tens of millions of dollars' worth of subsidies to farmers (Steinberg &

Tan, 2024). The revenue generated by tariffs for China during the trade-war would have been a temporary boost, but it came at the cost of higher prices for Chinese consumers. Small countries are price taker, but large country can influence the world prices through trade policy. For example, China impose tariffs on dairy can shift lower the world price from due to reduced demand and increased domestic supply (see appendix figure A11-A10). To generate the same level of tariff revenue (cdef) in a large country as that produced by the initial tariff T0 in a small country, the tariff should be higher (see figure 2) (Gaigné & Gouel, 2022; Schmitz et al., 2022).

Discussion

China is the largest importer of alfalfa, and the U.S. has comparative advantage in alfalfa trade with China because alfalfa is a land intensive crop ("China," 2021; Hou et al., 2024). China's demand for U.S. alfalfa remains strong because per capita farmland scarcity and diet structure changes determines the import demand for dairy products and forage (China, 2021). But ongoing tariffs, non-tariff barriers, and currency fluctuations continue to create headwinds for U.S. exporters. From 2020 to 2022, there is a sharp increase in U.S. forage exports due to China's limited resources for high quality alfalfa, and Chinese dairy consumption demand increased significantly (see appendix figure A8). The U.S. dairy prices experienced a significant decrease after 2022, due to alfalfa export reduced and inputs cost decreased (see appendix A6-A12). Changes in Chinese dietary structure, especially the growing demand for dairy products, highlight the importance of the alfalfa trade.

Tariffs imposed on higher elastic commodity have lower welfare effects on trade (Steinberg & Tan, 2024). An alfalfa tariff pushed up dairy marginal costs, shifting the dairy supply curve inward. Tariff impacts on dairy import quantities are greater than those on alfalfa, as China's demand for U.S. alfalfa is relatively more inelastic, whereas dairy imports are sourced from a more diverse set of countries. This upstream tax therefore propagates downstream, cutting dairy supply and raising dead-weight loss, while also harming alfalfa growers and consumers via higher effective costs. In short, given dairy's elastic demand and scale, we must pay special attention to alfalfa tariff pass-through, as it depresses dairy output and lowers social welfare. Because the U.S. is not a

major dairy supplier to China, tariffs on U.S. dairy can be largely offset by switching to Australia, New Zealand and EU sources, whereas China remains high demand of alfalfa from U.S., limiting substitution on the feed side.

Empirical results align with the large-country model's terms-of-trade channel in the context of the U.S.–China dairy–forage trade. Tariffs on both inputs (alfalfa) and outputs (dairy) compound cost pressures. Input tariffs increase the marginal cost of production, shifting the Chinese dairy supply curve upward, while output tariffs lower the effective demand for U.S. dairy, shifting down the world prices. So Chinese small dairy farms faced higher risk of costs exceeding prices. These findings align with Unveren and Luckstead (2020) and Farrokhi and Pellegrina (2023), who demonstrated that tariffs on agricultural inputs lead to reduced productivity, particularly in sectors reliant on imported resources. As table 1 and 2 illustration, it indicated that alfalfa and alfalfa elasticity of imports is inelastic in the short run, almost all cost of the tariffs was transferred to Chinese consumers and dairy farms. This is aligned with Amiti et al. (2019) study results of U.S. 2018 tariff impacts on price and welfare. Trump's tariff policy failed to account for the differing terms-of-trade effects that tariffs have on large versus small countries. The imposition of retaliatory tariffs result in input costs increasing, the small producer will get higher risk and consumer will afford transferred cost.

Three policy implications emerge from the analysis. First, trade diversification reduces vulnerability: expanding partnerships (e.g., alfalfa sourcing from EU and dairy sourcing from Australia) can raise the effective export supply elasticity σ_a, reducing the cross-market pass-through. Second, domestic production subsidies for inelastic input factors (Proposition **??**) are more cost-effective than equivalent output subsidies, because they directly address the binding cost channel. Third, the quality gap in alfalfa production between China and the United States—52.60 million tons of high-quality alfalfa hay in the U.S. versus 1.8 million tons in China in 2015 wang$_c hinas_2 020 - - -$
$underscorestheneedforcomplementaryinvestmentsinland, seeds, andirrigationtoimprovedomesticforageyields$
$al.(2014), whoviewtariffsasalegitimateinstrumentforaddressingchronictradeimbalances. UndertheScaledTari$

This study's twin-tariff framework has two main limitations. First, the U.S.–China channel is

not fully representative of China's dairy import system: New Zealand is China's dominant dairy supplier, while the U.S. plays 3rd. role on the output side even though it is the primary source of China's alfalfa hay. Meanwhile, alfalfa pellets/meal are sourced more broadly (e.g., Spain and Italy). Second, alfalfa is a core forage that supports milk protein and output volume and account largest share of dairy farm feeds cost, but dairy diets are typically multi-input bundles (e.g., alfalfa, silage, soybeans, oats) that vary by herd, and growth/lactation stage. As a result, modeling the feed shock primarily through alfalfa may not fully capture substitution across feed ingredients or adjustment along other margins. Therefore, we caution against over interpreting this result because this study provides a narrow view of the twin tariff. The insights of this study are applicable under twin-tariff regimes where inputs exhibit lower elasticity (owing to scarce substitutes and structural supply-chain rigidities), while import demand for output is elastic (as alternative foreign suppliers are comparatively accessible), analogous adjustment patterns and policy transmission effects may be expected (Wang & Zou, 2020; Wang et al., 2010). Even with these constraints, the paper contributes by linking international trade theory on tariff incidence and input–output intersection to an agricultural empirical setting, offering a tractable identification empirical pathway for studying twin tariffs and motivating future extensions.

Conclusion

Under U.S.-China trade wars dispute, China imposed tariffs on alfalfa that increased dairy production costs. This paper provides a study of the impacts of trade-war on livestock feed and dairy. The findings of this study illuminate the multifaceted impacts of U.S.-China trade wars on China's dairy sector. Increased tariffs on alfalfa and other feeds disrupted the input supply chain for dairy farmers in China. These challenges initially increased the marginal input cost of dairy production, and consequently, small-scale farms faced a higher risk of being squeezed out of the market because they are price takers. In trade between large economies, twin tariffs may push up the prices of inelastic inputs in the short run, while exerting negligible influence on elastic output prices in the long run. So twin tariff effects also related with substitute elasticity of products imposed on tariffs. Subsidies

for local forage and investment support for regional dairy have collectively improved the sector's resilience. Furthermore, increased tariffs and rising import costs also led China to ramp up policies supporting local production. These policies provide financial incentives for dairy production and forage production and improving breeding practices to boost output and quality.

U.S. alfalfa and soybean exports fell to a three-year low due to retaliatory tariffs,, even exports to other countries increasing (Adjemian et al., 2019; Morgan et al., 2022). The Chinese government implemented a series of measures aimed at bolstering domestic production, which included subsidies for alfalfa farmers, and identification of alternative imports partners. The study provides valuable insights for other international trade markets navigating similar trade challenges. With inelastic input factor demand, twin tariffs raise input cost, squeeze small producers, and increase social-welfare losses more than a comparable single-tariff regime. Twin tariffs generate higher dead-weight loss, redistribute surplus away from consumers and small producers, and could reduce long-run productive capacity through alternative effects and import shifts. In addition, frequent changes in tariff policy generate uncertainty and alter expectations for domestic forage traders and dairy producers, often triggering large-scale stockpiling or abrupt adjustments to production plans, which ultimately undermine the intended policy objectives. These unanticipated shifts are difficult to quantify and may disrupt supply chains, leading to pronounced price volatility driven by supply–demand imbalances. Moving forward, optimal tariff policies will be pivotal in ensuring the comparative competitiveness of agricultural trade.

References

Adenäuer, M., Frezal, C., & Chatzopoulos, T. (2025). Weathering the storm: Trade openness and the resilience of agricultural markets under increasing weather extremes. *Journal of Agricultural Economics*, *76*(2), 241–250. https://doi.org/10.1111/1477-9552.12635

Adjemian, M. K., Arita, S., Breneman, V., Johansson, R., & Williams, R. (2019). Tariff retaliation weakened the U.S. soybean basis. *Choices*, *34*(4), 1–9.

Amiti, M., Redding, S. J., & Weinstein, D. E. (2019). The impact of the 2018 tariffs on prices and welfare. *The Journal of Economic Perspectives*, *33*(4), 187–210. https://www.jstor.org/stable/26796842

Bohara, A. K., & Kaempfer, W. H. (1991). A test of tariff endogeneity in the United States. *The American Economic Review*, *81*(4), 952–960. https://www.jstor.org/stable/2006655

Bu, D., & Liao, Y. (2022). Land property rights and rural enterprise growth: Evidence from land titling reform in China. *Journal of Development Economics*, *157*, 102853. https://doi.org/10.1016/j.jdeveco.2022.102853

Cantor, A., Turley, B., Ross, C. C., & Glass, M. (2022). Changes to california alfalfa production and perceptions during the 2011–2017 drought. *The Professional Geographer*, *74*(4), 628–641. https://doi.org/10.1080/00330124.2022.2075409

China: China extends section 301 tariff exclusions on 12 agricultural products | USDA foreign agricultural service [USDA-FAS]. (2023, December 29). https://fas.usda.gov/data/china-china-extends-section-301-tariff-exclusions-12-agricultural-products

China: Market Overview - Alfalfa Hay and Other Forages | USDA Foreign Agricultural Service. (2021, December). Retrieved November 19, 2024, from https://fas.usda.gov/data/china-market-overview-alfalfa-hay-and-other-forages

Chor, D., & Li, B. (2024). Illuminating the effects of the US-China tariff war on China's economy. *Journal of International Economics*, *150*, 103926. https://doi.org/10.1016/j.jinteco.2024.103926

Fajgelbaum, P., Goldberg, P., Kennedy, P., Khandelwal, A., & Taglioni, D. (2024). The US-China trade war and global reallocations. *American Economic Review: Insights*, *6*(2), 295–312. https://doi.org/10.1257/aeri.20230094

Fajgelbaum, P. D., & Khandelwal, A. K. (2022). The economic impacts of the US–China trade war. *Annual Review of Economics*, *14*(1), 205–228. https://doi.org/10.1146/annurev-economics-051420-110410

Farrokhi, F., & Pellegrina, H. S. (2023). Trade, technology, and agricultural productivity. *Journal of Political Economy*, *131*(9), 2509–2555. https://doi.org/10.1086/724319

Felxi, B. (2024). *USDA ERS - countries & regions* [U.S. Department of Agriculture (USDA)]. https://www.ers.usda.gov/topics/international-markets-u-s-trade/countries-regions/

Fertő, I., Bakucs, Z., & Fałkowski, J. (2021). Dairy sector trade dynamics: Some insights on the evolution of trade linkages within the EU. *Journal of Agricultural Economics*, *72*(3), 698–711. https://doi.org/10.1111/1477-9552.12429

Gaigné, C., & Gouel, C. (2022, January 1). Chapter 88 - trade in agricultural and food products. In C. B. Barrett & D. R. Just (Eds.), *Handbook of agricultural economics* (pp. 4845–4931, Vol. 6). Elsevier. https://doi.org/10.1016/bs.hesagr.2022.03.004

Gooch, E. (2017). *China dairy supply and demand* (Economic Research Service). USDA-ERS. www.ers.usda.gov

Grant, J. H., Arita, S., Emlinger, C., Johansson, R., & Xie, C. (2021). Agricultural exports and retaliatory trade actions: An empirical assessment of the 2018/2019 trade conflict. *Applied Economic Perspectives and Policy*, *43*(2), 619–640. https://doi.org/10.1002/aepp.13138

Hanrahan, R. (2024, November 12). *Ag producers bracing for potential China trade war* [Farm policy news]. https://farmpolicynews.illinois.edu/2024/11/ag-producers-bracing-for-potential-china-trade-war/

Hou, P., Shen, B., Kong, X., Sawadgo, W., & Li, W. (2024). 2024 q1 agricultural trade series: US-China trade trends and opportunities for alfalfa exports. *Plains Press*. https://plainspress.

scholasticahq.com/article/124556-2024-q1-agricultural-trade-series-us-china-trade-trends-and-opportunities-for-alfalfa-exports

Interesse, G. (2024, November 11). *China-Mexico economic relations: Trade, investment, and opportunities* [China briefing news]. https://www.china-briefing.com/news/china-mexico-economic-relations-trade-investment-and-opportunities/

Jeff Luckstead & Stephen Devadoss. (2025). Consequences of trump's bilateral trade policies for agriculture. *Choices*, *40*(4), 13–18. https://doi.org/10.22004/AG.ECON.370424

Kondaridze, M., & Luckstead, J. (2023). Determinants of dairy-product trade: Do subsidies matter? *Journal of Agricultural Economics*, *74*(3), 857–873. https://doi.org/10.1111/1477-9552.12536

Koppenberg, M., & Hirsch, S. (2022). Output market power and firm characteristics in dairy processing: Evidence from three EU countries. *Journal of Agricultural Economics*, *73*(2), 490–517. https://doi.org/10.1111/1477-9552.12460

Li, M., Balistreri, E. J., & Zhang, W. (2020). The U.S.–China trade war: Tariff data and general equilibrium analysis. *Journal of Asian Economics*, *69*, 101216. https://doi.org/10.1016/j.asieco.2020.101216

Morgan, S., Shawn, A., Jayson, B., & Saquib, A. (2022, January). *The economic impacts of retaliatory tariffs on U.S. agriculture* (ERR No. 304). U.S. Department of Agriculture, Economic Research Service. www.ers.usda.gov

Nunn, N., & Trefler, D. (2010). The structure of tariffs and long-term growth. *American Economic Journal: Macroeconomics*, *2*(4), 158–194. https://doi.org/10.1257/mac.2.4.158

Qian, Z. (2023, December 25). *China import export tariffs in 2024* [China briefing news]. Retrieved November 19, 2024, from https://www.china-briefing.com/news/china-import-export-tariffs-in-2024/

Richman, J., Richman, H., & Richman, R. (2014). Balanced Trade: Ending the Unbearable Costs of America's Trade Deficits. *Political Science & Geography Faculty Books*. https://digitalcommons.odu.edu/politicalscience_geography_books/33

Rose, A. (2024, September 11). *Americans have been paying tariffs on imports from China for decades | PIIE* [Section: PIIE Charts]. https://www.piie.com/research/piie-charts/2024/americans-have-been-paying-tariffs-imports-china-decades

Sall, I., Tronstad, R., & Chin, C. Y. (2023). Alfalfa export and water use estimates for individual states. *Western Economics Forum*. https://doi.org/10.22004/ag.econ.337173

Schmitz, A., Moss, C. B., Schmitz, T. G., Kooten, G. C. v., & Schmitz, H. C. (2022, April 27). *Agricultural policy, agribusiness, and rent-seeking behaviour, third edition* (3rd. edition). University of Toronto Press.

Smith, V. H., & Glauber, J. W. (2020). Trade, policy, and food security. *Agricultural Economics*, *51*(1), 159–171. https://doi.org/10.1111/agec.12547

Steinberg, D., & Tan, Y. (2024). *How will China respond to biden's tariffs? look at trump's trade war. | PIIE* [Section: RealTime Economics]. https://www.piie.com/blogs/realtime-economics/2024/how-will-china-respond-bidens-tariffs-look-trumps-trade-war

Tucker, J. J., Mullenix, M. K., Rios, E., Basigalup, D., & Bouton, J. H. (2024). Systems management strategies for increasing alfalfa use in warm-humid regions. *Grassland Research*, *3*(2), 187–198. https://doi.org/10.1002/glr2.12080

Unveren, H., & Luckstead, J. (2020). Comprehensive broiler supply chain model with vertical and horizontal linkages: Impact of US–china trade war and USMCA. *Journal of Agricultural and Applied Economics*, *52*(3), 368–384. https://doi.org/10.1017/aae.2020.5

USDA ERS - agricultural trade [U.S. Department of Agriculture (USDA)]. (2024). https://www.ers.usda.gov/data-products/ag-and-food-statistics-charting-the-essentials/agricultural-trade/

USDA-FAS. (2023, January 6). *Record U.S. FY 2022 agricultural exports to China - USDA foreign agricultural service* [U.S. Department of Agriculture (USDA)]. https://fas.usda.gov/data/record-us-fy-2022-agricultural-exports-china

Victoria, D. (2024). *The economic impacts of retaliatory tariffs on U.S. agriculture | economic research service* [U.S. Department of Agriculture (USDA)]. https://www.ers.usda.gov/publications/pub-details?pubid=102979

Villoria, N. B. (2026). Trade frictions and domestic food price stability in the presence of large-scale climate shocks. *American Journal of Agricultural Economics*, *108*(1), 285–308. https://doi.org/10.1111/ajae.12531

Wan, W., Li, Y., & Li, H. (2022). Yield and quality of alfalfa (medicago sativa l.) in response to fertilizer application in china: A meta-analysis. *Frontiers in Plant Science*, *13*. https://doi.org/10.3389/fpls.2022.1051725

Wang, Q., Parsons, R., & Zhang, G. (2010). China's dairy markets: Trends, disparities, and implications for trade. *China Agricultural Economic Review*, *2*(3), 356–371. https://doi.org/10.1108/17561371011078462

Wang, Q., & Zou, Y. (2020). China's alfalfa market and imports: Development, trends, and potential impacts of the U.S.–China trade dispute and retaliations. *Journal of Integrative Agriculture*, *19*(4), 1149–1158. https://doi.org/10.1016/S2095-3119(19)62832-7

Wolff, A. W. (2024, May). *Trump's proposed blanket tariffs would risk a global trade war | PIIE* [Section: RealTime Economics]. https://www.piie.com/blogs/realtime-economics/2024/trumps-proposed-blanket-tariffs-would-risk-global-trade-war

Xu, Y., Elbakidze, L., Gassman, P. W., Yen, H., Ma, Y., Hubbart, J. A., & Arnold, J. G. (2025). U.S.–china agricultural trade and environmental outcomes: The case of nutrient runoff to the gulf of mexico. *Applied Economic Perspectives and Policy*, *47*(3), 983–1002. https://doi.org/10.1002/aepp.13501

Zhang, Z., & Abler, D. (2023). Dairy pricing policy, production, and water quality: Application to the chesapeake bay watershed. *Journal of the Agricultural and Applied Economics Association*, *2*(2), 350–365. https://doi.org/10.1002/jaa2.62

Appendix A. - Supplymental Figures and Tables

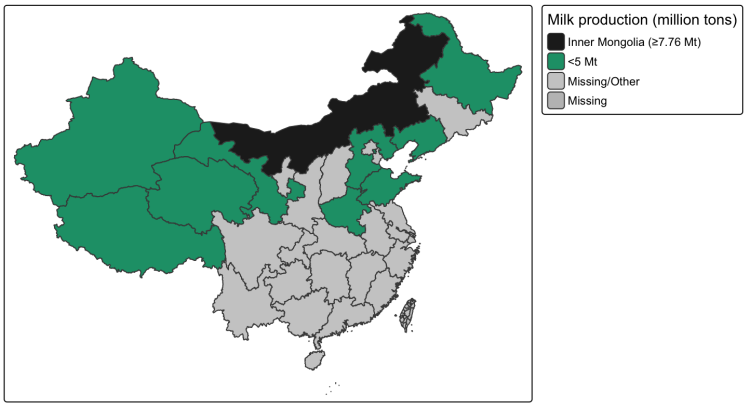

Figure A1: Chinese dairy production distribution

Source: China Food Consumption Report 2022

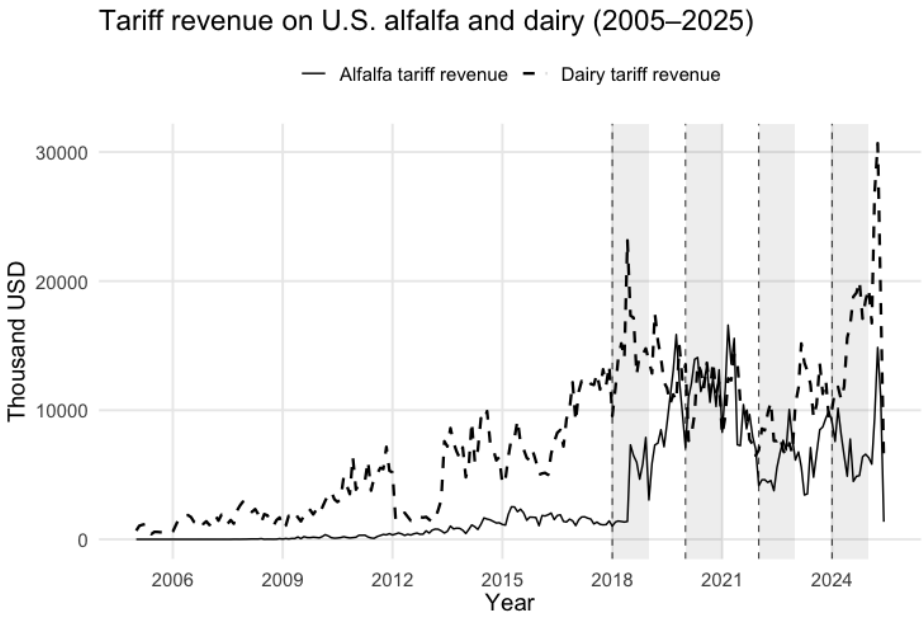

Figure A2: China's tariff revenue on U.S. dairy and alfalfa from 2005 to 2025

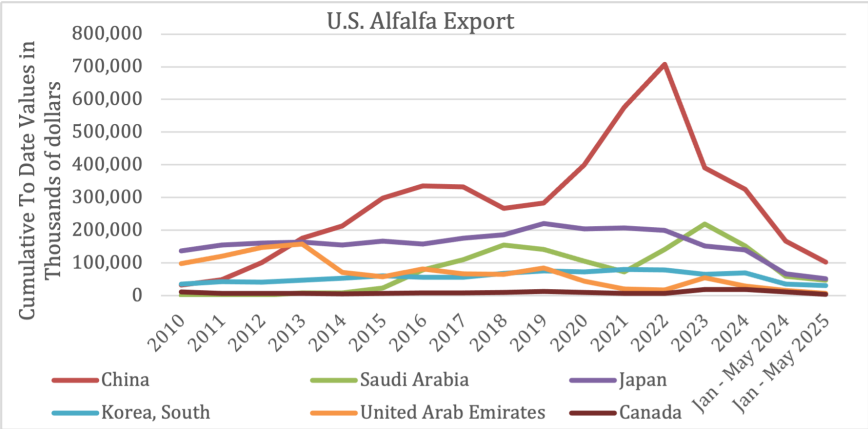

Figure A3: U.S. alfalfa export value from 2010 to 2025

Source: U.S. Census Bureau Trade Data

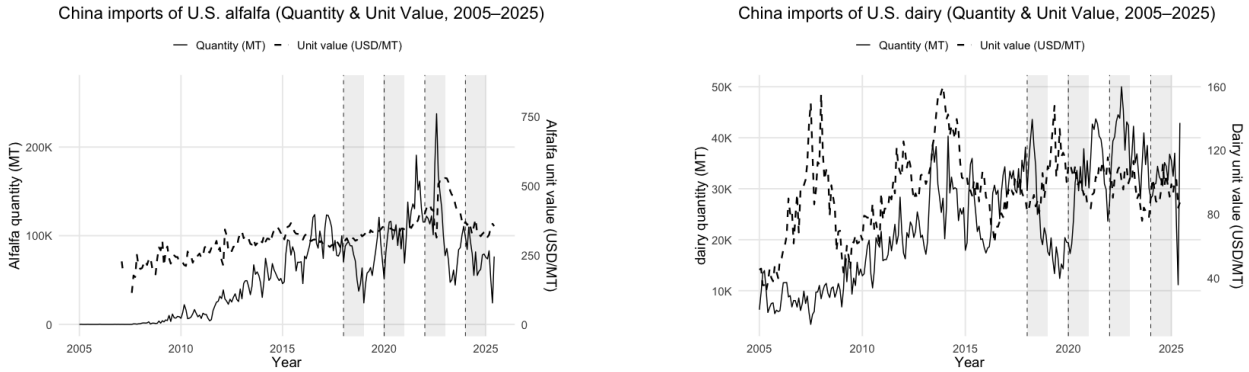

Figure A4: China import alfalfa (left) and dairy (right) quantity and price from U.S.

Source: GACC Statistics; USDA-FAS-GATS

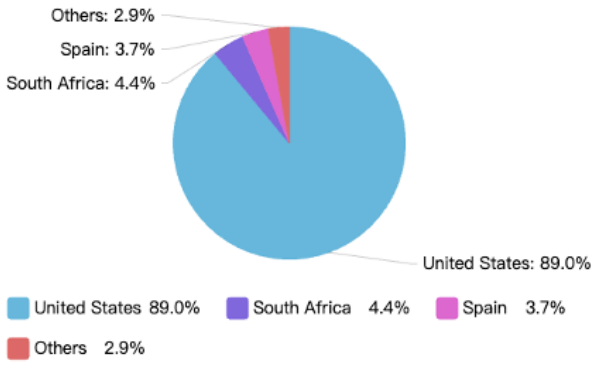

Figure A5: Chinese alfalfa hay imports source in 2023 (tons)

Source: 2024 Market Overview - Alfalfa Hay and Other Forages in People's Republic of China

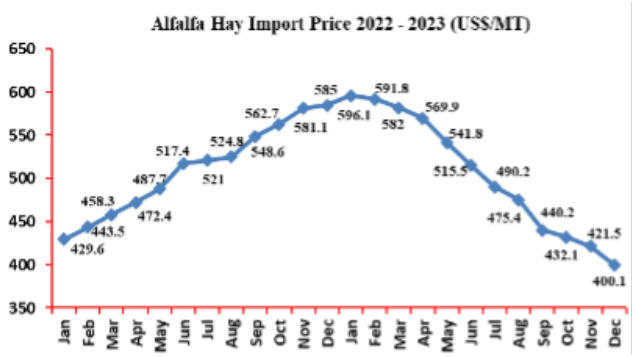

Figure A6: The price of China import alfalfa from U.S.

Source: USDA-FAS

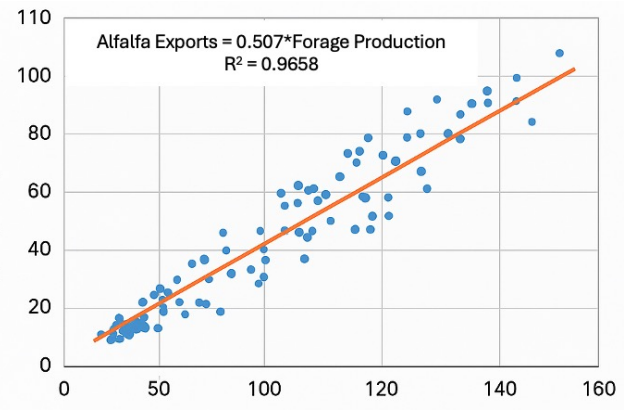

Figure A7: Relationship of U.S. alfalfa exports to all states to forage production exports (Million $)

Source: (Sall et al., 2023)

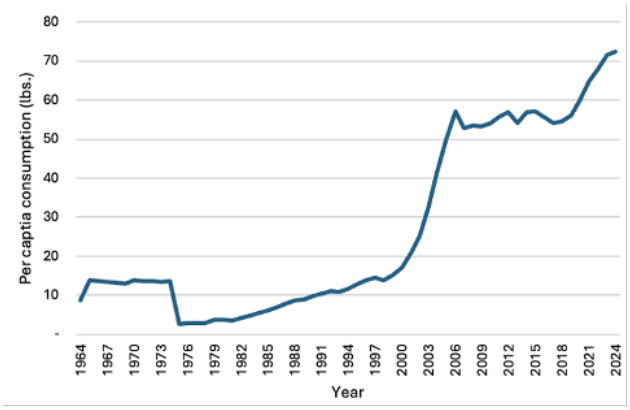

Figure A8: China per capital dairy consumption from 1964 to 2024

Source: Hoard's Dairyman Intel

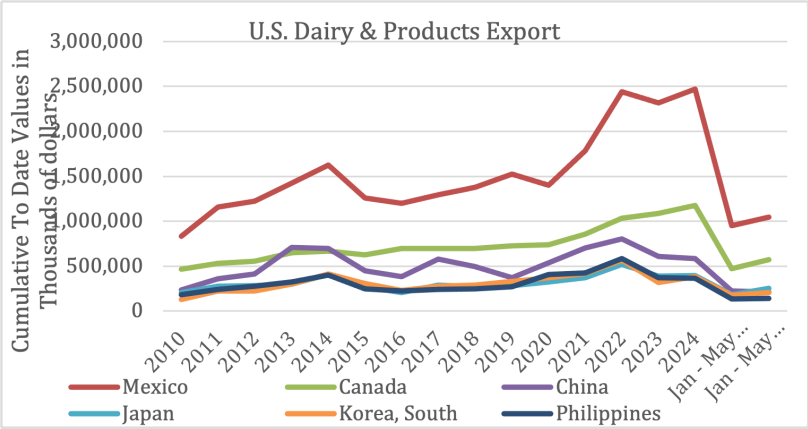

Figure A9: U.S. dairy & products export value from 2010 to 2025

Source: U.S. Census Bureau Trade Data

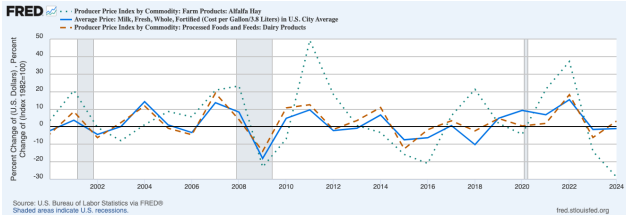

Figure A10: Average price of milk, fresh, whole, fortified in U.S. city (US$/gallon)

Source: U.S. Census Bureau Trade Data

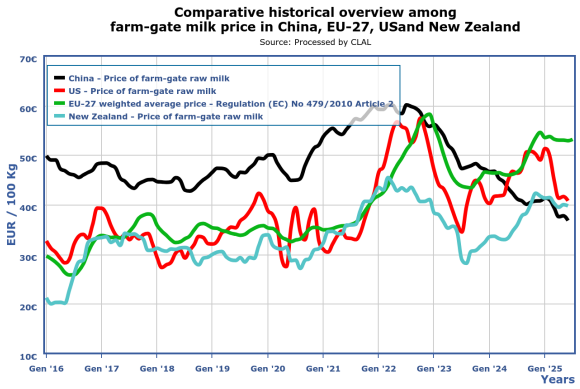

Figure A11: Farm gate milk price from 2016 to 2025

Source: Processed by CLAL, National Bureau Statistic

Figure A12: Annual dairy value losses due to retaliatory tariffs (Million USD)

Source: (Morgan et al., 2022)

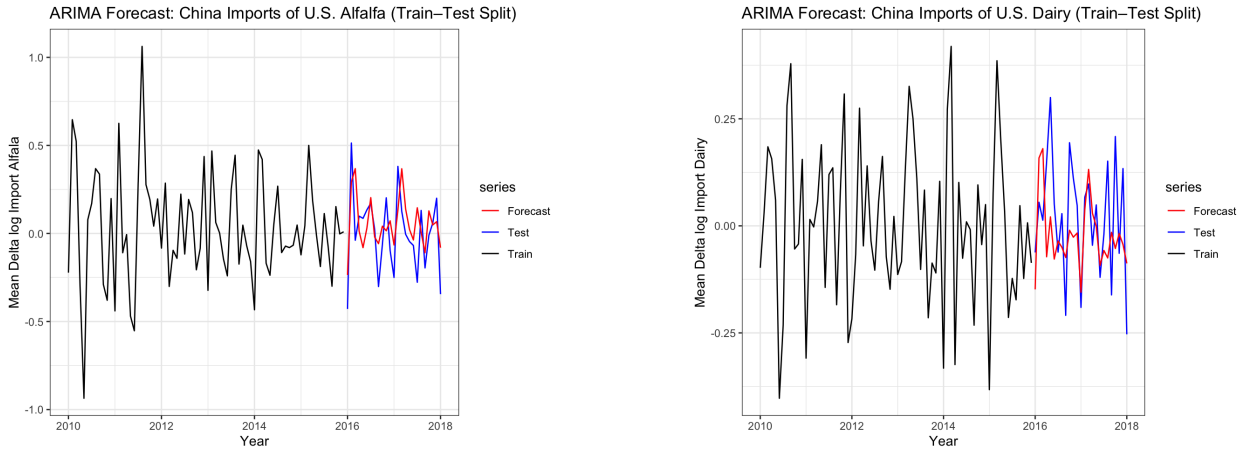

Figure A13: China import alfalfa (left) and dairy (right) quantity from U.S. stationary check

Table A1: Summary of cumulative twin-tariff effects on key variables

Outcome variable	Cumulative effect	Std. error	p-value	Observations	Adj. R^2
$\Delta\log$(Dairy import quantity)	-1.411^{***}	0.157	0.000	248	0.413
$\Delta\log$(Alfalfa import quantity)	-0.699	0.483	0.148	248	0.385
$\Delta\log$(Alfalfa import price)	0.093^{**}	0.036	0.010	248	0.594
$\Delta\log$(Farm-gate milk price)	0.022	0.031	0.443	248	0.971

Notes: Entries report cumulative 0–2 month effects from the preferred distributed-lag specification, calculated as the sum of the contemporaneous. The twin-tariff measure is defined as the sum of $\log(1+tr^{dairy})$ and $\log(1+tr^{alfalfa})$, where tr denotes the tariff rate in decimal form. Newey–West heteroskedasticity- and autocorrelation-consistent standard errors are reported. Significance levels: $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

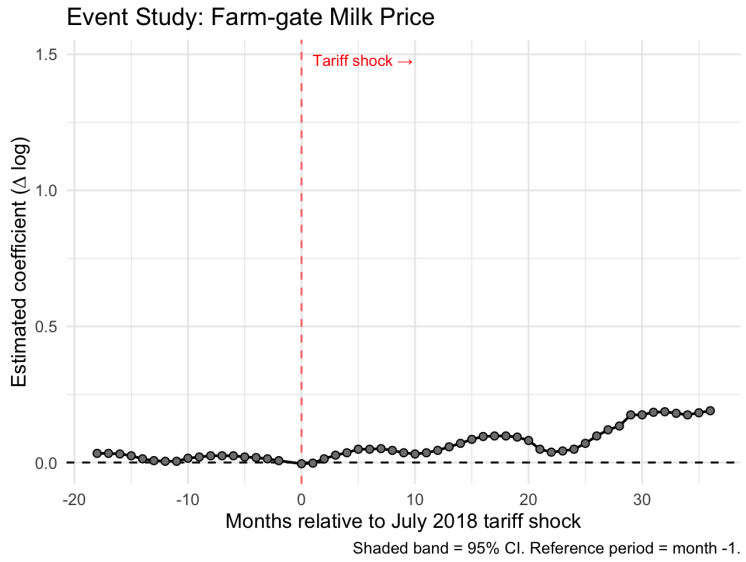

Figure A14: Tariff shock event impacts on China domestic milk price

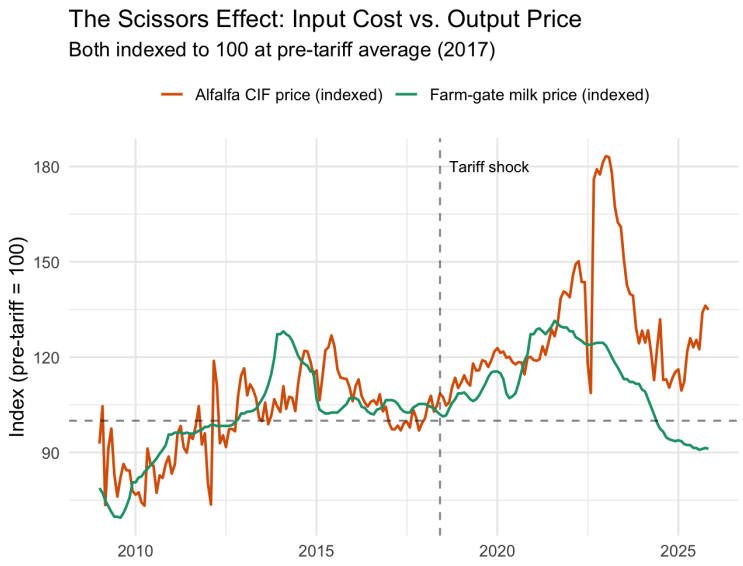

Figure A15: Alfalfa and farm-gate milk price before and post tariff war

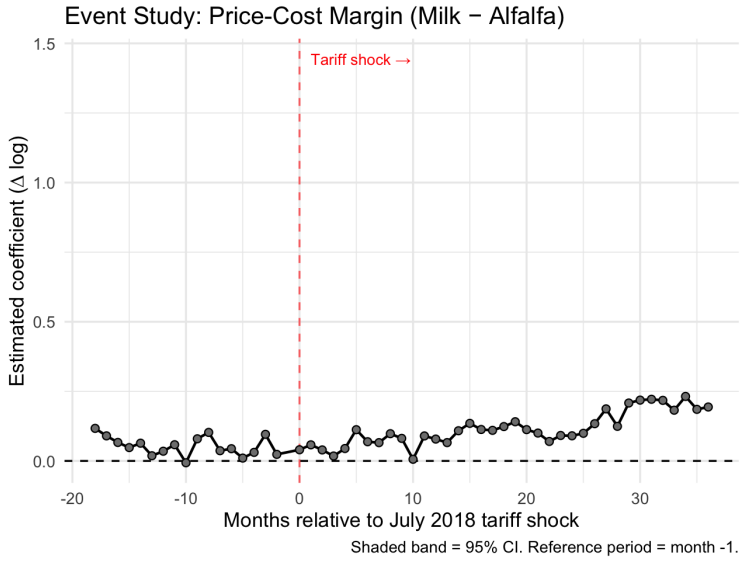

Figure A16: Margin price-cost

Appendix B. - Robustness Checks

Twin Tariffs Price Effects Check

Table B1: Robustness checks of twin-tariff price effects

Variable	(1) Baseline	(2) Interaction	(3) Margin	(4) FAO price proxy
$\Delta\ln(trd_t)$	0.0576	0.0079	-0.4409	0.0690
$\Delta\ln(trd_{t-1})$	-0.0138	-0.0065	0.3373	-0.0222
$\Delta\ln(trd_{t-2})$	-0.0346	-0.0042	-0.3909	-0.0351
$\Delta\ln(tra_t)$	0.0519	0.0114^{**}	-0.8346	-0.0450
$\Delta\ln(tra_{t-1})$	0.0043	0.0029	0.1447	0.0027
$\Delta\ln(tra_{t-2})$	0.0296	0.0106	0.1666	0.0398
$\Delta\ln(dairy_qty)$	-0.0073^{*}			
COVID	-0.0131^{***}	-0.0104^{*}	-0.0026	
$\Delta\ln(\text{fuel})$	0.0234^{*}	0.0229^{*}	0.0171	
$\Delta\ln(tra_t)\times\Delta\ln(trd_t)$		0.0712^{*}		
$\Delta\ln(faodp)$				-0.1165^{*}
$\Delta\ln(tra_t)\times\Delta\ln(faodp)$				0.7460
Observations	248	248	248	243
Adj. R^2	0.9713	0.9821	0.8314	0.7931

Notes: This table reports robustness checks for the estimated price effects of the twin-tariff shock on China's dairy production system. Column (1) presents the baseline reduced-form model for farm-gate milk price changes. Column (2) adds the contemporaneous interaction between dairy and alfalfa tariff changes. Column (3) uses the milk price–alfalfa price margin as the dependent variable. Column (4) includes a global dairy price proxy to account for external price movements and possible trade diversion. Heteroskedasticity-robust standard errors are used in all specifications. Significance levels: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Placebo Tests

First, we implement two placebo exercises to evaluate whether tariff shocks capture pre-existing trends. (i) We construct a placebo tariff series by shifting actual tariff changes three years earlier:

$$\mathrm{tr}_t^{\mathrm{placebo}} = \mathrm{tr}_{t+36}.$$

Re-estimating both the first-stage and second-stage equations with placebo tariffs yields coefficients that are statistically close to zero, suggesting that tariff shocks do not proxy for underlying trends. (ii) We augment the model by including *lead* tariff changes, $\Delta \ln(1+\mathrm{tr}_{t+1})$ and $\Delta \ln(1+\mathrm{tr}_{t+2})$. Consistent with exogeneity, the lead coefficients are near zero, indicating that future policy announcements do not predict past import volumes or domestic milk prices.

Alternative Lag Structures

To examine sensitivity to dynamic adjustment assumptions, we estimate variants of the baseline model with alternative lag lengths: (i) a shorter window (lags 0–1), (ii) a longer window (0–3 and 0–4), (iii) a geometric distributed lag, and (iv) a three-month average tariff index. Magnitude varies modestly across specifications, but the qualitative conclusions remain the same: tariff increases consistently reduce import quantities, with effects accumulating over the first two months.

Tariff-Quantity Decomposition

We further test robustness by decomposing tariff transmission into quantity and price channels. (i) We replace tariff rates with import unit values for dairy and alfalfa, $\Delta \ln P_t^D$ and $\Delta \ln P_t^A$, to isolate price pass-through. (ii) We include both tariff rates and CIF prices jointly in the first stage to test whether tariffs retain predictive power for import quantities after controlling for unit-value movements. Across all decompositions, tariff coefficients remain stable in sign and direction, confirming that results are not artifacts.

Additional Sensitivity Analyses

We also conduct supplemental checks, including excluding the melamine crisis years (2007–2009) and the COVID-19 period (2020–2021), and estimating models in levels rather than log-differences. All robustness exercises yield consistent qualitative patterns, strengthening confidence in the identification strategy.

Table B2: Summary of Robustness Checks

Robustness Exercise	**Description and Expected Outcome**
Placebo Tariff Shocks	Shift tariff series three years earlier; placebo coefficients close zero when tariffs are exogenous.
Lead Tariff Tests	Include future tariff changes; leads have no predictive power for past outcomes.
Short Lag Window (0–1)	Tests immediate adjustment; results align with baseline direction.
Long Lag Window (0–3, 0–4)	Captures delayed contracting/shipping effects; cumulative effects remain similar.
Geometric Distributed Lag	Smooth decay in tariff effects; sign and significance match baseline.
Averaged Tariff Index	Three-month tariff average; stabilizes noise, but effects remain consistent.
Tariff vs. CIF Price Decomposition	Replace tariffs with import prices; tests price pass-through robustness.
Excluding Crisis Years	Removes 2007–2009 and 2020–2021; results remain stable.

Appendix C. - Data Description

Tariff data time line

The 2018 U.S.-China trade war introduced a complex phenomenon, in which we define as "twin tariffs"- the simultaneous imposition of duties on upstream production inputs and downstream final products. In July 2018 China imposed an additional 25% retaliatory tariff on U.S. agricultural production include alfalfa, and most U.S. dairy products also became subject to substantial retaliatory duties (Xu et al., 2025). In 2019, many cheese lines faced totals around 33–37%, reflecting China's MFN tariff plus the 25% Section 301 retaliation. China announced another round in September 2019, adding 5% or 10% on selected U.S. goods; for dairy, products such as fresh cheese and grated cheese reached total applied tariffs of about 38–42%, while buttermilk and fermented milk could reach about 50% depending on the line. After the Phase One Agreement was signed in January 2020 and entered into force on February 14, 2020, China did not fully remove retaliatory tariffs, but it opened an exclusion process for affected farm goods and expanded market access for U.S. dairy facilities and forage products, including alfalfa hay pellets and cubes; in practice, this meant many dairy tariffs formally remained, but some products obtained exclusions. From 2021 to 2025, China continued to maintain retaliatory Section 301 tariffs on most U.S. dairy products, although key feed-related exceptions were repeatedly extended; notably, whey for feed use received repeated exclusions through February 28, 2025, and alfalfa had earlier benefited from exclusion treatment beginning in September 2019, which helped ease pressure on China's dairy-feed system even though the broader retaliatory tariff regime largely stayed in place. In May 2025, following the Geneva negotiations, China temporarily reduced the additional retaliatory tariff imposed during the 2025 escalation from 125% to 10% for 90 days.

Earlier tariff layers and product-specific measures were not necessarily eliminated in full. So the effective tariff burden on individual agricultural goods could still differ across products. In the Chinese dairy sector, province-level dairy price data are unavailable, but farm-gate milk prices are relatively uniform because production is concentrated in northern provinces and raw milk is

primarily purchased by large company such as Mengniu and Yili. In addition, forage tariffs are uniform across the relevant HS6 products, while dairy tariffs vary across HS6 lines. However, most high-variation products, such as butter, lactose, and cheese, represent only a small share of Chinese consumption; the main imported dairy products (milk powder, liquid milk, and whey) face the same tariff rate. Together, these institutional features support the use of the twin-tariff framework as a reasonable sector-level approximation.

Data coding and source

All data process coding are open access on author's GitHub and sole responsibility of the authors: https://github.com/liyoumin/Trade-tarrifs-impacts-on_Chinnese_dairy/blob/main/estimation.R

Table C1: Data sources and variable descriptions.

Variable	**Source**	**Code**	**Notes**	**Time**
U.S. export dairy value	USDA FAS (GATS)	HS04&0410	$ Thousand	01/2005–06/2025
U.S. export alfalfa value	USDA FAS (GATS)	HS1214	$ Thousand	01/2005–06/2025
U.S. export alfalfa quantity	USDA FAS (GATS)	HS1214	Metric ton	01/2005–06/2025
U.S. export dairy quantity	USDA FAS (GATS)	HS04&0410	Metric ton	01/2005–06/2025
FAO dairy price index	FAO (weighted avg. 2014–2016)	HS04&0410	Index	01/2005–07/2025
China farm-gate milk price	CLAL.it	—	CNY / kg	01/2009–06/2025
China imports forage grass	GACC (China Customs)	HS12	Metric ton	01/2010–06/2024
Feed & fodder quantity	USDA FAS (GATS)	0120AT	Metric ton	01/2010–06/2024
U.S. alfalfa price	USDA NASS	—	$ / ton	2010–2024
U.S. dairy price	BLS via FRED	—	% change	1995–2025
U.S. grazing fee	USDA NASS	—	$ / animal / month	2010–2024
Tariff rate	Chad P. Bown (2025 Chart)	—	—	06/2018–03/2025
Tax revenue (state/local)	U.S. Census Bureau	—	Million $	01/2005–12/2024
Tariff rate (ISIC)	ISIC	—	%	01/2005–12/2024
Ag. export value	USDA ERS	HS04&12	$ Thousand	01/2005–06/2025